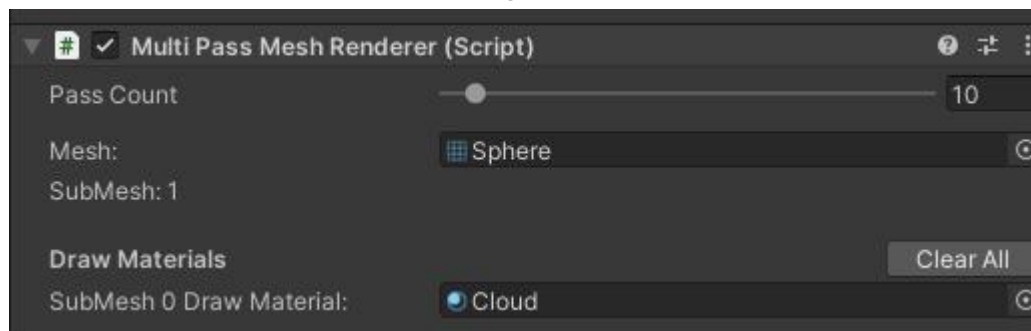


Get Started

MeshRenderer

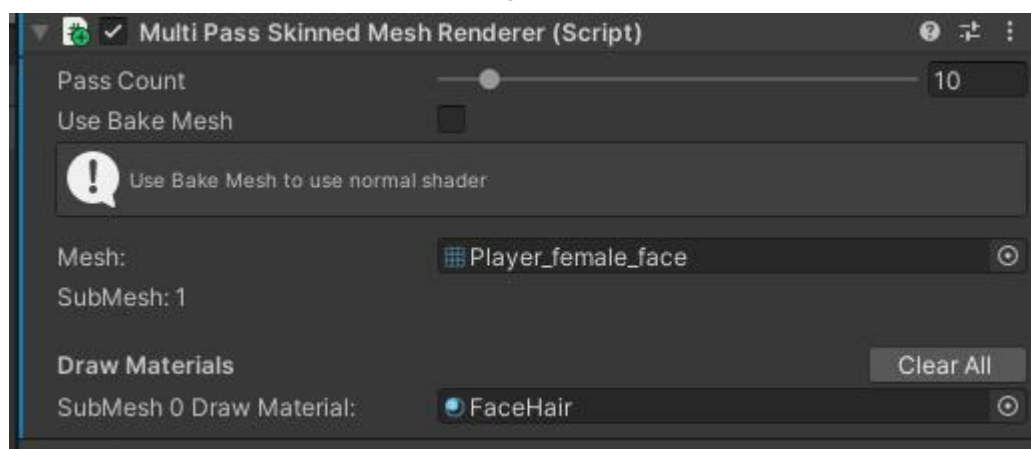
1. Add Component "MultiPassMeshRenderer"
2. Assign Material to "SubMesh x DrawMaterial".
3. For example you can find "Fur.mat" or "Cloud.mat" in "MultiPassFX/Scene/Materials"
4. Material must set "Enable GPU Instancing"



5. Material in MeshRenderer will also rendered, If you want to hide it, replace it with "HideSubMesh.mat"

SkinnedMeshRenderer

1. Add Component "MultiPassSkinnedMeshRenderer"
2. Assign Material to "SubMesh x DrawMaterial".
3. For example you can find "Fur.mat" or "Cloud.mat" in "MultiPassFX/Scene/Materials"
4. Material must set "Enable GPU Instancing"



5. Material in SkinnedMeshRenderer will also rendered, If you want to hide it, replace it with "HideSubMesh.mat"
6. For Devices don't support OpenGL ES 3.1+AEP, will auto fallback to "UseBakeMesh"

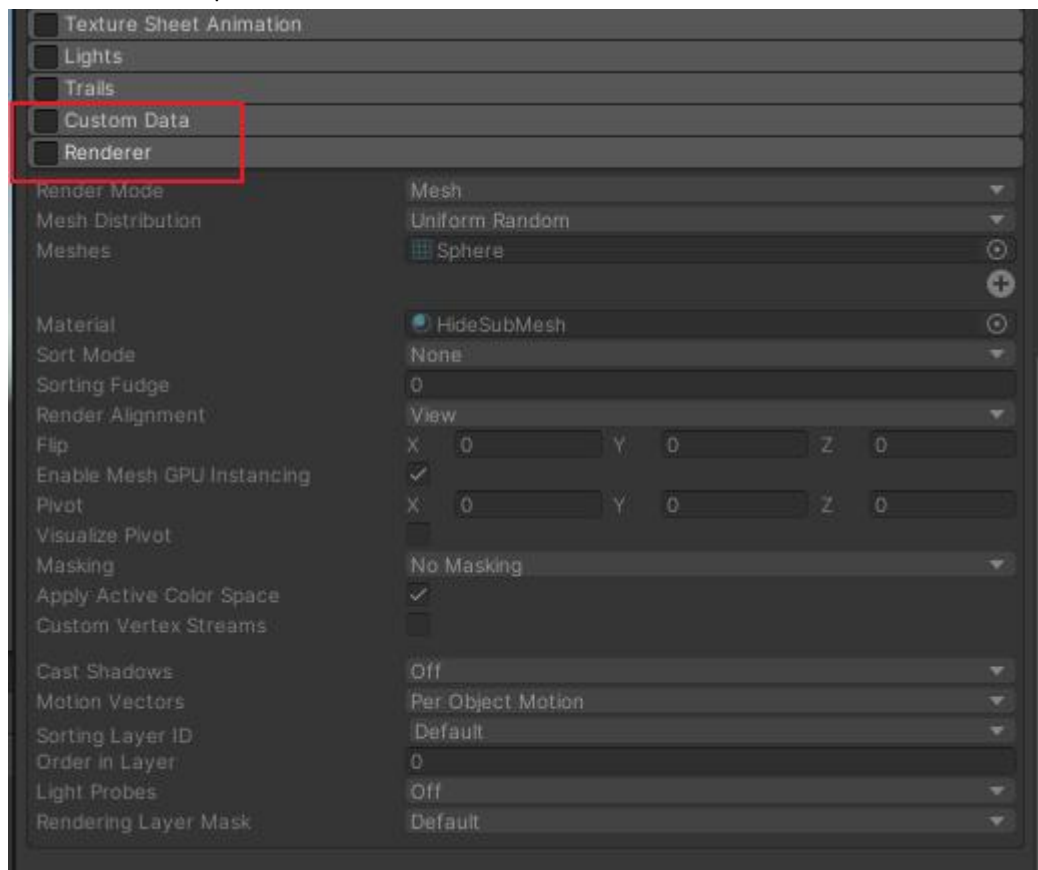
7. If "UseBakeMesh" not set, we will use GraphicsBuffer to render mesh after skinning, so we must use modified shader. See Shaders in "MultiPassFX/Shaders" for more information
8. When refreshing Unity, this component maybe invisible, just disable and enable it and you can see it again. It's a UnityEditor bug.

ParticleSystem

1. Add Component "MultiPassParticleRenderer"
2. Assign Material to "DrawMaterial"



3. For example you can find "Fur.mat" or "Cloud.mat" in "MultiPassFX/Scene/Materials"
4. Material must set "Enable GPU Instancing"
5. Note: This component will auto set "Renderer" module disable

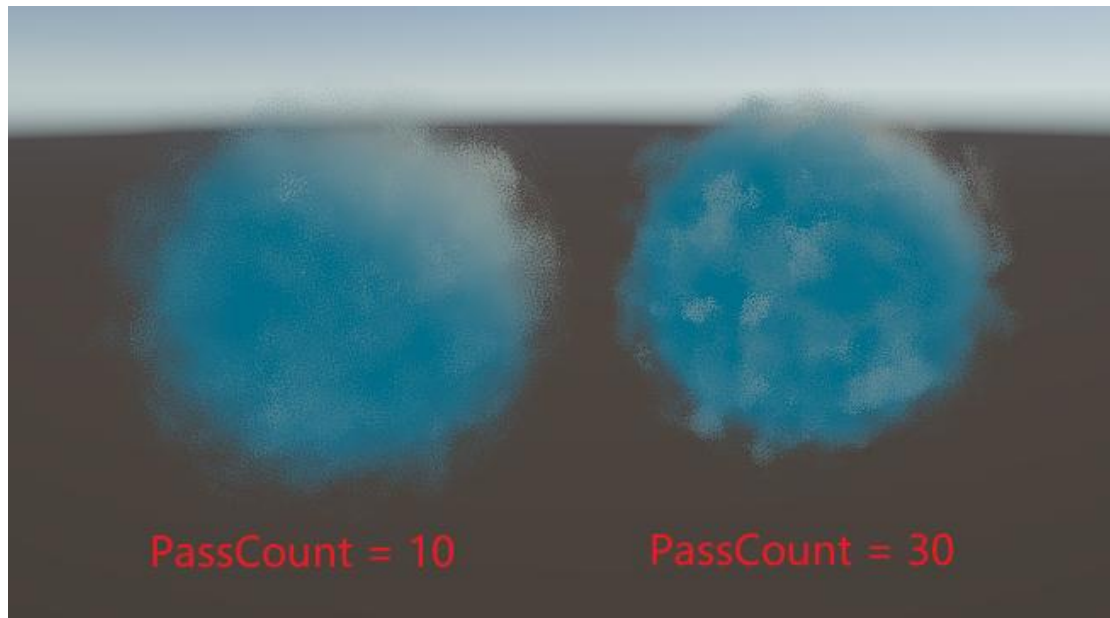
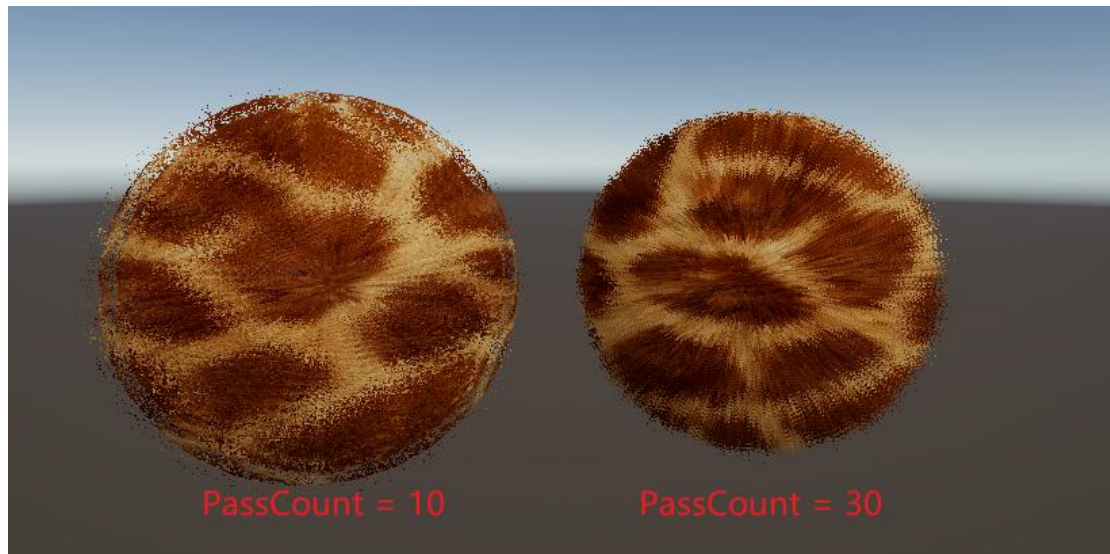


Demo Scene

You can find demo scene in “MultiPassFX/Scene/SampleScene.unity”, in scene there is all usage of MultiPassFX

Details

Pass Count: Large for better visual, But cost more gpu time



Customize Skinned Shader

If you want make your own shader support GraphicsBuffer for SkinnedMeshRender, follow these steps:

1. Include "MultiPassFX/Shaders/SkinnedMeshBuffer.hlsl"
2. Use "SKINNED_MESH_VERTEX_INPUT" in vertex input (Contain position / normal / tangent)

```
#include "SkinnedMeshBuffer.hlsl"

struct Attributes
{
    SKINNED_MESH_VERTEX_INPUT
    float2 uv : TEXCOORD0;
    UNITY_VERTEX_INPUT_INSTANCE_ID
};
```

3. Add "SETUP_SKINNED_MESH_INPUT(input)" in VertexShader

```
float _DepthFadeDistance;

Varyings CloudVert(Attributes input)
{
    Varyings output;

    UNITY_SETUP_INSTANCE_ID(input);
    UNITY_TRANSFER_INSTANCE_ID(input, output);
    SETUP_SKINNED_MESH_INPUT(input);

    VertexNormalInputs normalInput = GetVertexNormalInputs(input.normalOS, input.tangentOS);
    output.tangentWS = normalInput.tangentWS;
    output.bitangentWS = normalInput.bitangentWS;
    output.normalWS = normalInput.normalWS;
```

Optimizations

Reduce "PassCount" when object is far from camera